Case Study 1 - Atari

Student submission

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Worksheet link

https://courses.ethereallab.app/assignment/v4/IT265-001-F2026/case-study-1-atari/ad2374

Grading link

https://courses.ethereallab.app/assignment/v4/IT265-001-F2026/case-study-1-atari/grading/ad2374

AI disclosure: Completed; 1 entry; 0 declared AI use; 1 declared no AI use; policy v1

Game Overview

Game details

Response prompt

Identify the game: title, developer, publisher, original platform, genre, and release date. Note the version you played if different.

Submitted response

Centipede was developed by Ed Logg and Dona Bailey and published by Atari, Inc. It was released in arcades on June 30, 1981. The game is an action shooter in which players control a Bug Blaster to shoot a centipede and other insects while avoiding enemies. I played the online AARP version of Atari's Centipede through the AARP Games website, rather than the original arcade cabinet. The original platform was the arcade, and the version I played was a browser-based adaptation.
https://www.museumofplay.org/games/centipede/
https://www.arcade-history.com/?id=427&n=centipede-upright-model&page=detail

History and industry

Response prompt

Explain the historical context of the release and where the game fit in the industry at that time.

Submitted response

Centipede was released during the golden age of arcade video games, a period when games such as Space Invaders (1978) and Pac-Man (1980) helped establish arcades as a major entertainment industry. Atari distinguished Centipede with its colorful insect theme, mushroom filled playfield, and trackball controls in the original arcade version. It became one of Atari's most commercially successful arcade games. Its accessible controls and distinctive visual design also helped it reach a broad audience during a period when arcade developers were exploring alternatives to traditional space themed shooters.
https://atari.com/pages/history
https://www.museumofplay.org/games/centipede/

Contemporary comparison

Response prompt

Compare the game with a similar game from the same period, explaining a meaningful design similarity or difference.

Submitted response

Centipede can be compared with Space Invaders (1978), another influential arcade shooter. Both games require players to shoot approaching enemies while avoiding threats and trying to achieve a high score. However, Space Invaders features rows of aliens moving across the screen, while Centipede features a segmented enemy that changes direction when it encounters mushrooms. These obstacles make the environment affect enemy movement, requiring players to adjust their aim and positioning. This gives Centipede a more dynamic interaction between enemies and the playfield while preserving the straightforward action of an arcade shooter.

https://www.museumofplay.org/games/centipede/
https://www.museumofplay.org/games/space-invaders/

Gameplay Mechanics And Level Design

Core gameplay mechanics

Response prompt

Explain how the core actions and rules shape a recurring player decision. Support the explanation with a gameplay example.

Submitted response

The core mechanics of Centipede create a recurring decision between attacking enemies and positioning the shooter safely. The player moves along the bottom of the screen and fires upward, while the centipede changes direction when it encounters mushrooms. Shooting a segment can create another mushroom and split the centipede into separate sections. For example, shooting the middle of a centipede can leave two groups approaching from different directions. The player must decide where to aim and when to reposition instead of firing without considering the consequences. This creates a balance between offense, positioning, and survival.
https://www.museumofplay.org/games/centipede/

Level design

Response prompt

Explain how the layout, obstacles, or progression affect a player decision or challenge. Use a specific example.

Submitted response

Centipede uses a fixed screen filled with mushrooms that function as obstacles and influence enemy movement. When the centipede encounters a mushroom, it changes direction and eventually moves farther down the screen. For example, a dense cluster of mushrooms can cause the centipede to reverse direction repeatedly, making its path harder to predict. The player must track these changes while watching for other insects that can enter the playfield. As the game progresses, faster movement and additional threats increase difficulty. This design makes the player manage both enemy behavior and the limited space available for movement.
https://www.museumofplay.org/games/centipede/

Graphic And Audio Design

Visual design

Response prompt

Explain how a visual design choice affects the experience or communicates information during play. Use a specific example.

Submitted response

Centipede uses colorful graphics and distinct enemy shapes to make the action easier to follow. The centipede's connected segments communicate its direction, while mushrooms are visually distinct obstacles. Spiders, fleas, and scorpions also have different appearances that help players recognize additional threats. For example, the scorpion can be identified as it moves across the playfield and poisons mushrooms, changing how those obstacles affect the centipede. The bright colors and unusual garden setting also give the game a distinctive appearance compared with the darker, space themed shooters popular at the time.
https://www.museumofplay.org/games/centipede/
https://spectrum.ieee.org/centipede-game

Sound design

Response prompt

Explain how music, sound effects, or their absence affect the experience or communicate information during play. Use a specific example.

Submitted response

The original arcade version of Centipede uses electronic sound effects to reinforce the action and create tension. Firing sounds provide feedback when the player shoots, while other effects accompany enemies and gameplay events. For example, when several insects approach the player's position, the combination of movement and firing sounds can make the action feel more urgent. Sound supports the game's fast-paced arcade experience rather than communicating a detailed story through dialogue or music. The browser version may reproduce or modify the original audio so its specific sound effects should be considered separately from the arcade version.
https://spectrum.ieee.org/centipede-game

Narrative And Storytelling

Narrative and setting

Response prompt

Explain how the setting and story relate to play. Describe key events if present, or explain the effect of a limited or absent story.

Submitted response

Centipede has a minimal narrative, with no substantial plot, dialogue, or story progression. The player defends against insects in a stylized garden, with the mushrooms and insect enemies forming the game's setting. This setting directly supports the gameplay because mushrooms change the centipede's movement, while spiders, fleas, and scorpions create additional threats. The player does not need to learn a complicated backstory to understand the objective of surviving and scoring points. The limited narrative keeps attention on the arcade action and makes the environment a functional part of the mechanics rather than a separate storytelling experience.
https://www.museumofplay.org/games/centipede/
https://atari.com/pages/centipede-was-one-of-the-first-with-a-significant-female-player-base

Character development

Response prompt

Explain how character development relates to gameplay, or how limited or absent development affects the experience.

Submitted response

Character development is basically absent from Centipede. The player controls a shooter, but the game does not develop its personality, background, or personal goals through a story. The centipede and other insects function primarily as gameplay threats rather than characters with individual motivations or character arcs. Instead of following a character's growth, players improve through practice, learning enemy movement, aiming more accurately, and positioning themselves more effectively. This lack of character development suits the arcade format because players can understand the objective immediately and focus on improving their performance.
https://www.museumofplay.org/games/centipede/

Impact And Reception

Reception

Response prompt

Summarize the critical and player reception using identifiable reviews or feedback. Distinguish release-era reception from later reactions.

Submitted response

Centipede became one of Atari's most commercially successful arcade games during the early 1980's. A contemporary 1983 review in Video magazine praised the arcade game's popularity and easy to use trackball controls, providing evidence of its positive release era reputation. Later reception has also been positive: the game was inducted into the

World Video Game Hall of Fame in 2020. These sources represent different forms of reception: the contemporary review reflects opinions close to the games original release, while the later induction recognizes its lasting historical significance. The evidence suggests that its accessible controls and distinctive gameplay helped it remain memorable long after its arcade debut.
https://www.ataricompendium.com/archives/magazines/video_review/video_review_apr83.pdf
https://www.museumofplay.org/games/centipede/

Industry and cultural impact

Response prompt

Explain the game's industry impact and cultural significance using research. Avoid claiming influence that the sources do not establish.

Submitted response

Centipede contributed to the success of arcade gaming during the industry's golden age and became one of Atari's best selling arcade titles. It was also notable for being co-designed by Dona Bailey, one of the few women working in arcade game development at the time. Its bright colors, garden setting, and trackball controls helped distinguish it from many contemporary shooters. Historical accounts identify Centipede as one of the early arcade games to attract a significant female player base, although this does not mean that all women preferred its design. Its induction into the World Video Game Hall of Fame in 2020 demonstrates its lasting recognition as an important part of video game history.
https://atari.com/pages/centipede-was-one-of-the-first-with-a-significant-female-player-base
https://spectrum.ieee.org/centipede-game
https://www.museumofplay.org/games/centipede/

Reflection

Fun and experience

Response prompt

Explain what made the game enjoyable or frustrating for you, using a gameplay example.

Submitted response

I chose Centipede for this case study because I have played it many times in the past and have always enjoyed its straightforward but challenging gameplay. My familiarity with the game made it interesting to analyze how its mechanics create a recurring challenge. For example, when the centipede splits into multiple sections, I have to adjust my aim and watch for threats approaching the bottom of the screen. Successfully destroying enemies is satisfying because it requires quick reactions and good positioning. However, the increasing speed and number of enemies can become frustrating when several threats approach at once. Revisiting a game I already enjoyed allowed me to look beyond the fun and better understand how its mechanics, level design, and difficulty contribute to the overall experience.
https://www.museumofplay.org/games/centipede/

Learning and research

Response prompt

Reflect on what this case study taught you and how easy or difficult it was to find dependable information.

Submitted response

This case study taught me that a game with simple rules can still create complex decisions through its mechanics and level design. In Centipede, mushrooms are not just decoration because they can change enemy movement, while shooting centipede segments can create new obstacles and additional threats. I also learned that visual design, sound effects, and limited narrative can all support a game's main objective. Researching the game's history showed me the importance of distinguishing the original arcade release from later versions and separating contemporary reviews from modern historical assessments. Information about its mechanics and historical significance was relatively easy to find, but determining which sources were reliable and which claims they supported required more care.
https://www.museumofplay.org/games/centipede/
https://spectrum.ieee.org/centipede-game